

BK Restaurant Management System - Technology Stack Finalization

Layer	Technology	Purpose	Justification
Architecture Pattern	Three-Tier Modular Architecture	Separation of layers	Improves maintainability and scalability
Frontend (Web)	React.js	POS, Dashboard, Inventory Management	Reusable and responsive UI
UI Framework	Tailwind CSS	Styling and responsive design	Modern and lightweight
Mobile App	Flutter	Delivery Rider Application	Cross-platform support
Backend	Node.js + Express.js	REST APIs and Business Logic	Scalable and efficient
Database	PostgreSQL	Store business data	Reliable relational database
Cache Layer	Redis	Caching and session management	Improves performance
Authentication	JWT	Secure authentication	Supports role-based access
Real-Time Communication	Socket.IO	Live updates	Real-time order and delivery tracking
Maps & GPS	Google Maps API	Navigation and tracking	Supports GPS delivery tracking
Notifications	Firebase Cloud Messaging	Push notifications	Reliable mobile notifications
Reporting & Charts	Chart.js / Recharts	Analytics dashboard	Data visualization
Hosting Platform	AWS / Railway.app	Application deployment	Scalable cloud hosting
Version Control	Git + GitHub	Source control	Team collaboration
CI/CD	GitHub Actions	Automated deployment	Continuous integration and delivery
Testing	Jest, RTL, Postman	Testing	Quality assurance
API Style	REST API	System communication	Standard API architecture